Recruitment Chatbot

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Project Features

1. Automated Applicant Screening for Resume Filtering using Artificial Intelligence

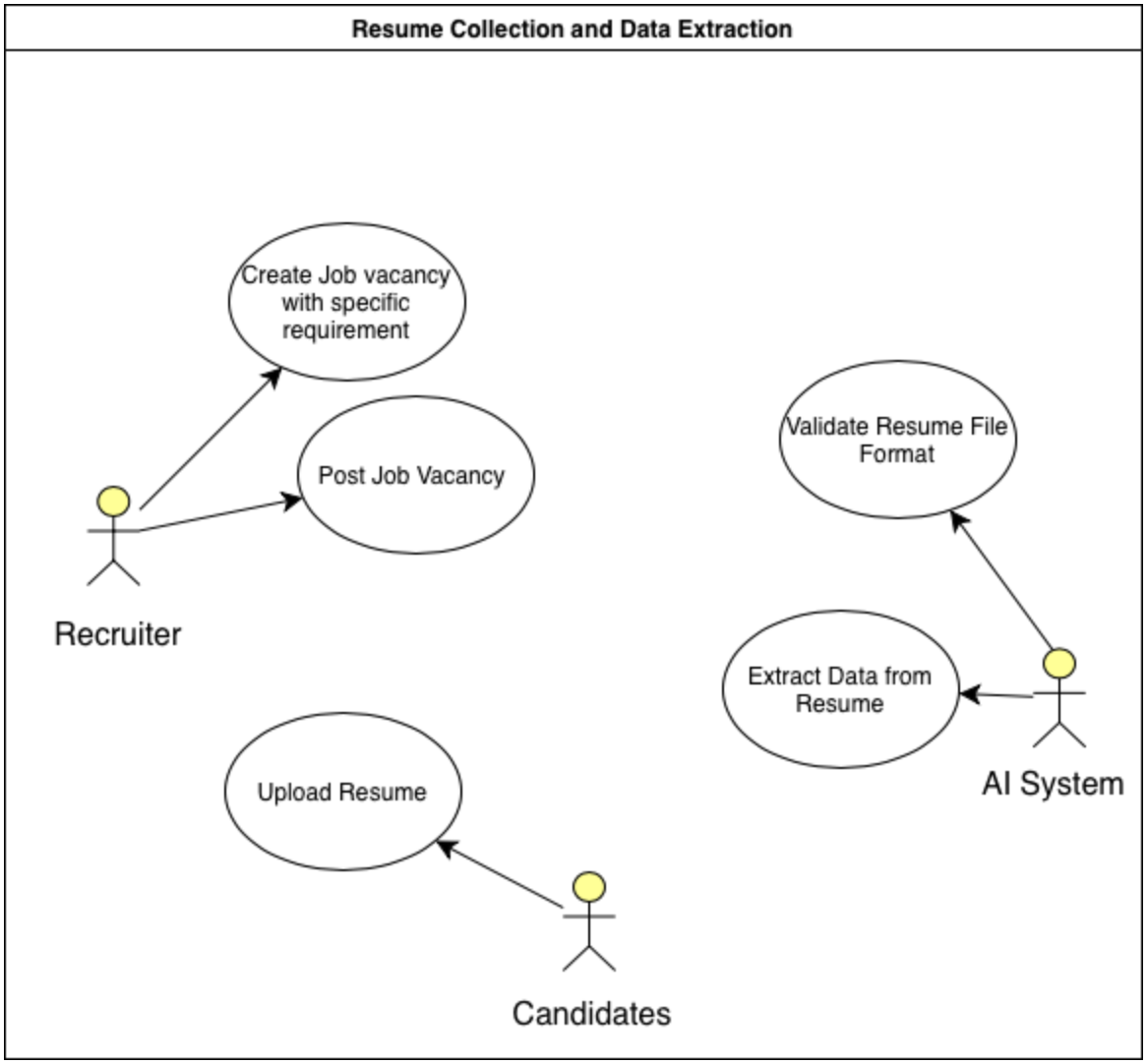
This feature uses Artificial Intelligence (AI) to automatically screen and filter applicant resumes without the need for full manual review. Instead of recruiters reading every CV, the system intelligently analyzes the content of each resume to determine whether a candidate meets the job requirements.

1.1. Business Process 1

Resume Collection and Data Extraction:

focuses on collecting candidate resumes and extracting relevant information from them in a structured, usable format.

1.1.1. Use Cases



Business Process Diagram (BPMN)

HTML link: https://modeler.camunda.io/share/33482475-6d40-4240-b38f-3641b9448e46
.PNG:

1.1.2. User Stories

Requirement ID	Epic	User Story Summary
REQ-RC&DE-1	Resume Collection and Data Extraction	As a candidate.I want to access the company's portal ,so that I can upload my resume.
REQ- RC&DE -2	Resume Collection and Data Extraction	As a candidate, I want to receive a confirmation mail that I've submitted my application ,so that I know it went through to the right party.
REQ-RC&DE-3	Resume Collection and Data Extraction	As an AI System, I want to learn historical data of previous successful candidates ,so that I can do accurate analysis.
REQ- RC&DE -4	Resume Collection and Data Extraction	As an AI System, I want to extract key information from candidates' resumes, so that I can ensure organized data analysis.
REQ-RC&DE-5	Resume Collection and Data Extraction	As an AI System, I want to categorize skills, education level and experience ,so that I can compare candidates fairly.
REQ- RC&DE -6	Resume Collection and Data Extraction	As an AI System, I want to flag the recruiter if a resume is incomplete or if there's any inaccuracies, so that the recruiter follows up with that candidate.
REQ- RC&DE -7	Resume Collection and Data Extraction	As a Recruiter, I want to receive a structured profile of each candidate ,so that reviewing applications would be faster
REQ- RC&DE -8	Resume Collection and Data Extraction	As a Recruiter, I want to be able to search and filter candidates upon extracted data ,so I can find suitable candidates efficiently.

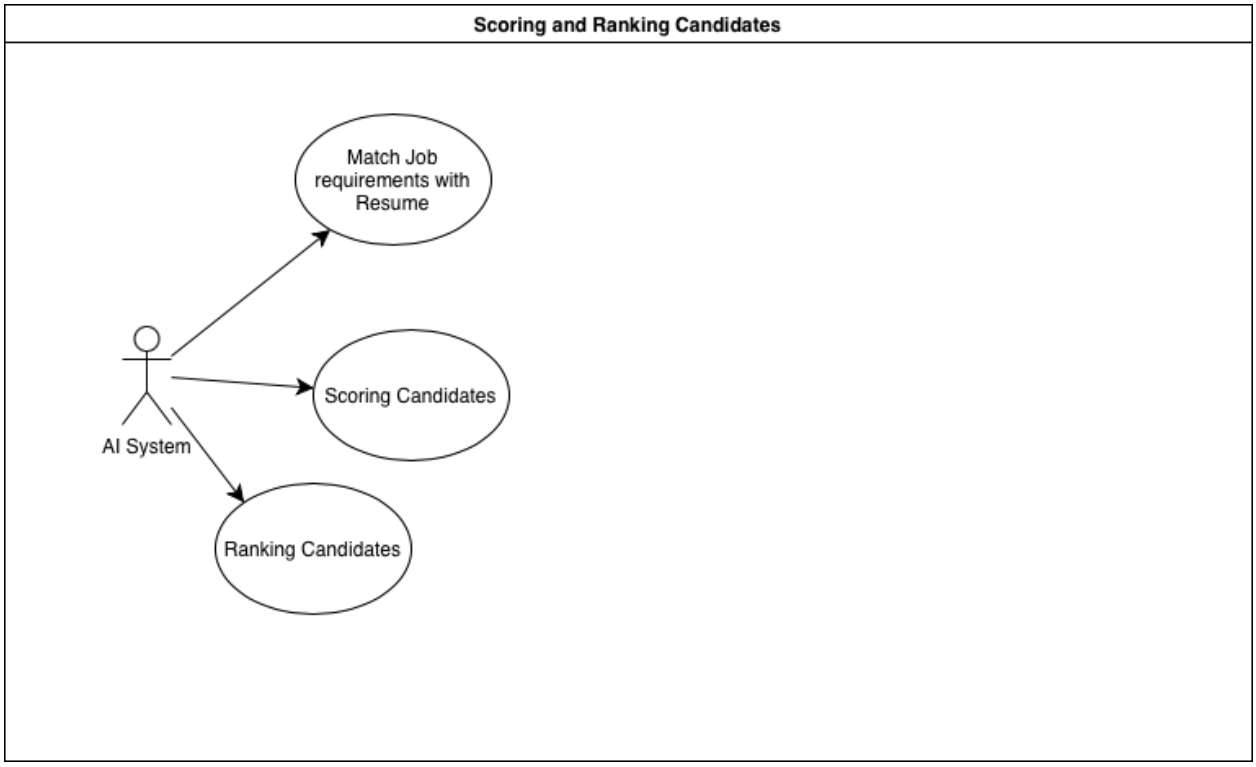
REQ- RC&DE -9	Resume Collection and Data Extraction	As a Recruiter, I want to post a job vacancy with its job requirements ,so that I can fulfill this role.
REQ-RC&DE-10	Resume Collection and Data Extraction	As an AI System, I want to validate the file format of Resume uploaded if it's PDF or Docs ,so that the right file can be submitted to extract from.
REQ-RC&DE -11	Resume Collection and Data Extraction	As a Recruiter, I want to create a job vacancy with specific job description and qualifications so I can fulfill the role in the company

1.2. Business Process 2

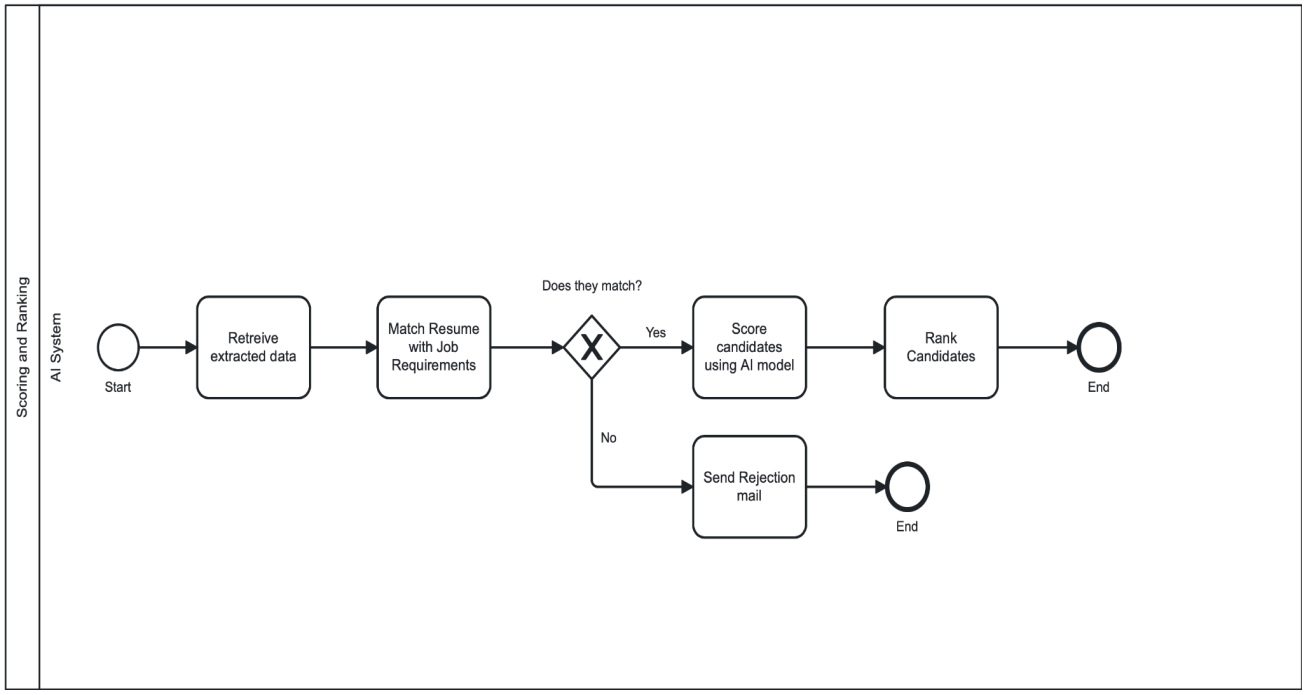
Scoring and Ranking Candidates:

focuses on evaluating candidate profiles and assigning scores or ranks based on how well they match the job requirements.

1.2.1. Use Cases



1.2.2. Business Process Diagram (BPMN)



1.2.3. User Stories

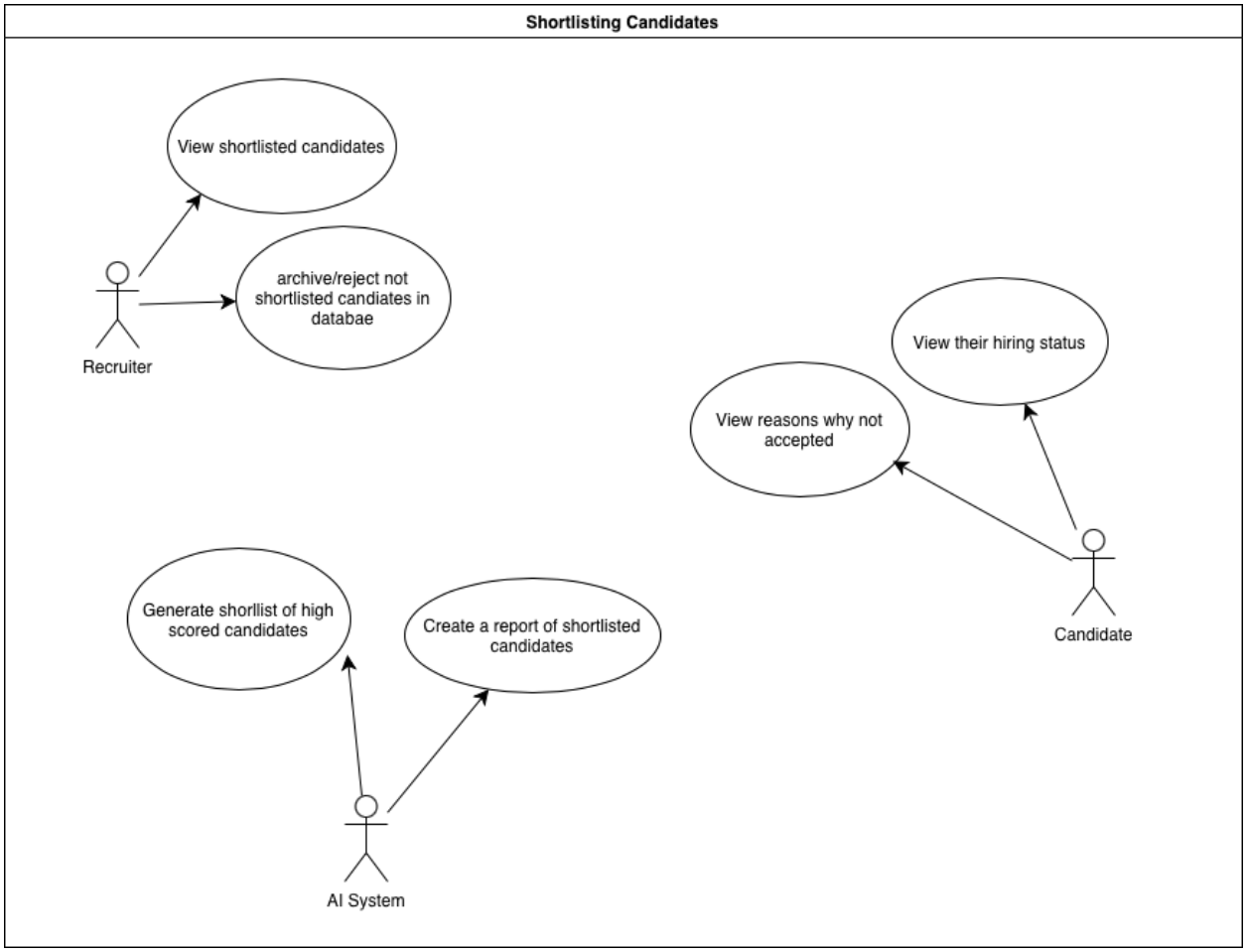
Requirement ID	Epic	User Story Summary
REQ-S&R-1	Scoring and Ranking Candidates	As a candidate .I want to be evaluated based on my qualifications ,so I can be evaluated fairly.
REQ-S&R-2	Scoring and Ranking Candidates	As an AI System, I want to score candidates based on education level,skills, professional experience and courses ,so that recruiters have an adequate list of suitable candidates.
REQ-S&R-3	Scoring and Ranking Candidates	As an AI System, I want to rank all candidates according to their scores ,so that I can prioritize top candidates.
REQ-S&R-4	Scoring and Ranking Candidates	As an AI System, I want to learn from recruiters' feedback, so that I can ensure accurate scoring.
REQ-S&R-5	Scoring and Ranking Candidates	As a Recruiter, I want to view all candidates' AI generated ,so that I can make better hiring decisions.
REQ-S&R-6	Scoring and Ranking Candidates	As a Recruiter, I want to update AI scores of candidates if inaccuracies occurred ,so that I can give subjective/human judgement.
REQ-S&R-7	Scoring and Ranking Candidates	As a Recruiter, I want to compare similar ranked candidates ,so that I can shortlist efficiently.
REQ-S&R-8	Scoring and Ranking Candidates	As an AI System, I want to match candidates' resumes with job requirements so that I guarantee adequate and suitable candidates.

1.3 Business Process 2

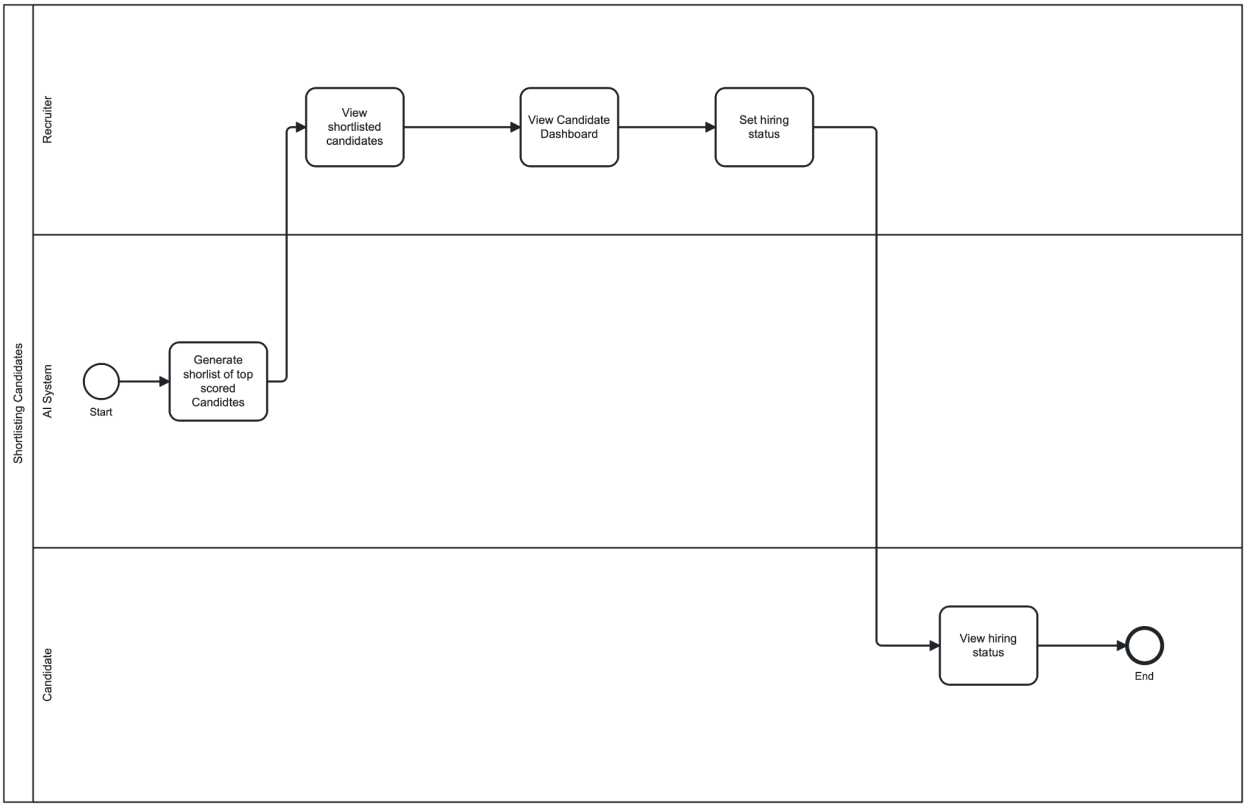
Shortlisting Candidates:

selecting a subset of candidates from the larger applicant pool who best meet the job requirements and it prepares a manageable list of high-potential candidates for recruiter review or interviews.

1.3.1. Use Cases



1.3.2. Business Process Diagram (BPMN)


Shortlisting Candidates
Recruiter
AI System
Candidate
Start
Generate shorlist of top scored Candidtes
View shortlisted candidates
View Candidate Dashboard
Set hiring status
View hiring status
End

1.1.3. User Stories

Requirement ID	Epic	User Story Summary
REQ-SC-1	Shortlisting Candidates	As a candidate .I want to be notified when shortlisted ,so I can prepare for the next hiring process.
REQ-SC-2	Shortlisting Candidates	As a candidate ,I want to receive feedback when not shortlisted ,so I can improve for future applications.
REQ-SC-3	Shortlisting Candidates	As an AI System, I want to generate a shortlist of qualified candidates based on scores ,so that the recruiter can view best fits only.
REQ-SC-4	Shortlisting Candidates	As an AI System, I want to provide feedback for shortlisted, so that I can ensure transparent hiring decisions.
REQ-SC5	Shortlisting Candidates	As a Recruiter, I want to review AI generated shortlists ,so that I can confirm or modify selected candidates.
REQ-SC-6	Shortlisting Candidates	As a Recruiter, I want to communicate with shortlisted candidates ,so that they can proceed with the upcoming step of hiring process.
REQ-SC-7	Shortlisting Candidates	As a Recruiter, I want to archive or reject not shortlisted candidates so that the database remains organized and ready for the next runner up candidate if the current one does not want to proceed.
REQ-S&R-8	Scoring and Ranking Candidates	As an AI System, I want to match candidates' resumes with job requirements so that I guarantee adequate and suitable candidates.

Field Requirements / Business Rules

Field & field requirements:

- **First Name** — Required, text only.
- **Last Name** — Required, text only.
- **Email Address** — Required, must be a valid email format.
- **Contact Number** — required, must be numeric/valid phone format.
- **Skills** — required; multi-select or tags; predefined list recommended.
- Education — Required ,must be selected from a predefined dropdown list:Diploma,Bachelor,Master,PhD
- **Experience** — Required; numeric years; dropdownlist; minimum number of years
- **Country** — Required; dropdown from predefined list.
- **Certifications** — Optional; free text or multi-select list; add extra score.
- **Languages** — Optional; multi-select list with proficiency levels.
- **Projects** — Optional; short description; add extra score.
- **Resume Upload** — Required; accepts PDF/DOCX only, size limits apply.
- **Cosine Similarity Resume-job match score: ≥0.7 Strong, 0.4–0.7 Medium, <0.4 Poor**

- Recruiter opens candidate profile, sees candidate info + uploaded CV + skills. Recruiter changes status to "Shortlisted," adds internal notes, schedules interviews (uses interview scheduling module/calendar integration), and logs that info. The candidate doesn't see internal notes or other candidates.

Wire frame

https://v0.app/chat/recruitment-suite-design-4-MAXhJHt4KCI?ref=66JJJ7

Test Cases:

Test Case ID	Test Scenario	Test Steps	Expected Result
RES 01	Upload supported file formats	1) Go to "Apply Job". 2) Upload resume in PDF, DOCX.	Upload successful. File stored.
RES-02	Upload unsupported formats	1) Upload PNG/JPG/ZIP.	System blocks upload: “File format is invalid”.
RES-03	Missing resume	1) Try submitting without uploading.	Validation error: “Resume is missing”.
AI-FMT-01	Detect incomplete resume	1) Upload resume missing key sections (ex: no experience).	AI marks Invalid Format and highlights missing sections.
AI-EXT-01	Extract basic fields	1) Upload resume.	AI extracts: full name, email, phone, education, experience.

MTCH-01	Match candidate to vacancy using TF-IDF	1) Upload resume. 2) The system converts resumes and job descriptions to TF-IDF vectors.	The system calculates cosine similarity and returns score (0–1).
MTCH-02	High match score	1) Resumes have the same skills as a job.	Similarity > 0.75. Mark as “value x100 % Match”.
MTCH-03	Feature mismatch detection	1) Candidate education < required education.	System reduces match score or marks candidate as Not Eligible.
AI-ShortList-01	Notify candidate when shortlisted	1) AI generates a shortlist. 2) Candidates are included. 3) The system sends notifications.	The candidate receives “We are glad to inform you, you have been shortlisted”.
AI-ShortList-02	Send feedback when not shortlisted	1) AI marks candidates as not shortlisted. 2) System sends feedback.	The candidate receives constructive feedback.
AI-ShortList-03	Generate shortlist based on scores	1) AI ranks candidates. 2) Filters top scores. 3) Displays shortlist.	The shortlist shows top qualified candidates.
AI-ShortList-04	Provide feedback for shortlisted candidates	1) AI creates shortlist. 2) AI prepares score breakdown. 3) Feedback shown.	The candidate sees feedback explaining shortlist reason.
Rec-ShortList-01	Review & modify AI shortlist	1) Open AI shortlist. 2) Recruiter edits list.	Updated shortlist saved.
Rec-ShortList-02	Communicate with shortlisted candidates	1) Open candidate. 2) Send message.	The candidate receives a message.

ARCH-01	Archive rejected candidates	1) Mark as Rejected. 2) System archives candidate.	The candidate moved to the Archive.
EVAL-01	Evaluate candidate fairly	1) Upload resume. 2) AI extracts data. 3) Score generated.	Fair score based on qualifications.
AI-SCORE-01	AI scoring based on education/skills/experience	1) Upload resumes. 2) AI calculates score.	Higher qualifications = higher score.
AI-RANK-01	Rank candidates automatically	1) Scoring completed. 2) System sorts by score.	Ranked list displayed.
AI-FEED-04	AI learns from recruiter feedback	1) Recruiter edits score. 2) AI stores feedback.	AI improves scoring accuracy.
Rec-SCORE-01	View AI-generated scores	1) Open candidate list. 2) View score column.	All AI scores visible.
Rec-SCORE-02	Update inaccurate AI scores	1) Select candidate. 2) Edit score.	Updated score saved.
Rec-RANK-01	Compare similarly ranked candidates	1) Open comparison panel. 2) Select candidates.	Comparison table displayed.
RES-04	Access portal to upload resume	1) Open careers portal. 2) Click Apply Now.	Portal loads successfully.
App-CNFRM-01	Receive confirmation on submission	1) Submit application. 2) Check email.	Confirmation email received.
AI-MDL-01	AI learns historical data	1) Upload historical data. 2) AI analyzes.	Scoring model updated.
SRCH-01	Search/filter candidates	1) Open list. 2) Filter by skills/experience.	Matching candidates displayed.

POST-01s	Post job vacancy	1) Go to Post Job. 2) Enter details. 3) Save.	Job vacancy created successfully.